Introduction to Data Science

Final Project — Instructions & Requirements

Semester 2, תשפ״ו (2025/2026)

The final project replaces the exam for this course. It is your opportunity to practice the full data science workflow: formulating a question, exploring and cleaning data, applying appropriate methods, and drawing sound conclusions. Expected effort is approximately 3 working days per person. You do not need to use every method covered in the course, but you should use a significant portion of it. You are welcome to use methods not covered in class (e.g., additional clustering or prediction algorithms), provided you explain the general idea behind them and why you chose them.

Submissions must be done in pairs — this project is not to be completed individually or in groups larger than two. Exceptions must be arranged with the lecturer in advance.

1. Timeline & Grade Weights

Milestone	Deadline	Weight
Project proposal	15/8/2026	20%
Final report + Jupyter notebook (code)	18/10/2026	40%
Final presentation (in scheduled meeting)	Within 2 weeks after 18/10/2026 (exact slot TBD)	40%

Missing a deadline without prior approval will be treated as a late submission per the course late-submission policy.

2. Project Proposal (20%)

Submit a short proposal (up to 2 pages) containing the following three parts:

2.1 Research question

State exactly what you plan to investigate. The question must be well-defined in the sense taught in this course: specific and answerable with data, not vague (e.g., "Who is the best player in the NBA?") and not trivial (e.g., "How many shows does Netflix release per category?").

A good question is interesting to a specific audience — the general public, or a defined professional/decision-making group (e.g., a transportation authority, a school district, a hospital, a policy maker) — and its answer could plausibly inform a real decision or understanding. Public/governmental datasets are often a good source of such questions, for example:

- Public transportation — e.g., what drives delays or ridership on a bus/train network, and where would added service help most
- Education — e.g., what factors are associated with student performance or dropout, using school/district-level data
- Public opinion / surveys — e.g., what predicts a particular attitude or voting pattern across demographic groups
- Health — e.g., what factors are associated with a health outcome or hospital readmission in a public health dataset
- Environment — e.g., what factors drive air/water quality or energy consumption across regions or over time
- Wellness — e.g., what lifestyle or demographic factors are associated with sleep, exercise, or mental well-being in a survey dataset
- Political — e.g., what factors are associated with voting patterns, policy positions, or election outcomes across regions or demographic groups

These are examples, not requirements — any domain is acceptable as long as the question is specific, answerable, and clearly matters to some real audience.

2.2 Dataset

Provide a link to the dataset, or attach it if it is not available online. Describe:

- Number of data points and number of features
- The columns relevant to your question
- The target column, if relevant (e.g., for prediction tasks)
- Anything else important to understanding the dataset

2.3 Methods

Outline the steps you plan to take and how you will carry them out, for example:

- Exploratory data analysis: checking for missing values and columns that look wrong, by plotting the data and comparing min/max values to the mean and standard deviation
- Looking for correlations between variables and the target variable
- Splitting the data into training / testing / validation sets (e.g., 50%-30%-20%)
- Building a model (e.g., linear regression) and evaluating it (R^2, significance, most contributing variables)
- Turning the target into a categorical variable under some criterion and applying a classifier (e.g., random forest), then evaluating it

Your plan should be tailored to your own project — the above is only an example. Things will likely change once you start implementing; the proposal's purpose is to confirm your question is answerable and that you know how to begin.

Note: the steps listed above are only an illustrative example. A more complex methodology than a single linear regression or a single classifier is expected.

Using a variety of appropriate tools is rewarded: for example, comparing more than one classifier, adding clustering to explore structure in the data, or examining feature importance, in addition to a single classification/regression model. This is not required, but a project that only ever runs one classifier and stops will score lower on project design than one that combines several complementary methods thoughtfully.

2.4 Proposal checklist

☐ Research question is specific, answerable, and non-trivial

☐ Dataset link/file included, with size, features, and target column described

☐ Planned methods listed step by step

☐ Proposal is 2 pages or less

2.5 Proposal grading policy

The proposal grade will be returned a few weeks after submission and is final — it will not be revised upward based on improvements made afterward in the final submission. Along with the grade, you will receive improvement recommendations as feedback.

Implementing those recommendations in your final report/notebook will not change the proposal grade itself, but not implementing them (without a good reason) can lower your final report grade, since the recommendations are meant to be incorporated into your finished project.

3. Final Report + Jupyter Notebook (40%)

Submit two components: (1) a written project report, and (2) the code and results as a single Jupyter notebook. You may submit multiple files as a single zip archive, or a notebook alone if it contains everything (in which case no zipping is required).

3.1 Report content

The report should be concise — its purpose is to let me understand what you did and why, not to be exhaustive. It must include:

- Background — a short description of the topic and the context needed to understand it
- Question — the question you answered. If it changed from your proposal, explain what changed and why
- Methods — the steps you took to answer the question, including the packages and parameters you used

- Results — a coherent description of your findings, with figures and tables (move very large tables to an appendix)
- Discussion — what your results mean, whether your approach was appropriate, what you would do differently, what remains unknown, and how you would address it. If you could not answer the question, explain why and what would be needed to answer it
- AI usage statement — see Section 5

3.2 Report format requirements

Requirement	Specification
Length	Approx. 3–5 pages, excluding figures, tables, and appendices
File format	PDF or Word (.docx)
Figures/tables	Embedded in the relevant section; move oversized tables to an appendix
Referencing your notebook	Where relevant, note which notebook cell/section produced a result

3.3 Jupyter notebook requirements

- Runs top-to-bottom without errors (Kernel → Restart & Run All) before submission
- Organized in the same order as the report (background/EDA → methods → results → discussion), using markdown cells as section headers
- Each non-trivial step has a short markdown or comment explaining what it does and why
- Cleaned, final version only — remove abandoned attempts and leftover debugging output
- All figures/tables referenced in the report are reproducible from the notebook

3.4 Submission checklist

☐ Report includes all five required sections plus the AI usage statement

☐ Notebook runs top-to-bottom with no errors

☐ Notebook is organized and commented, matching the report's structure

☐ Figures/tables in the report are legible and correctly labeled

☐ Files zipped together, or a single self-contained notebook submitted

4. Final Presentation (40%, part of the meeting)

Each student presents their project in a short, formal slide presentation during the scheduled meeting slot, followed by questions.

Element	Specification
Format	Formal slide deck (PowerPoint/Google Slides/PDF)
Length	Up to 10 minutes presentation, within a 30-minute meeting slot (remaining time for Q&A and discussion)
Slide count (guideline)	~8–10 slides
Required content	Motivation & question · dataset & methods (brief) · key results (1–2 main figures) · conclusions & limitations
Submit slides by	Before your assigned slot (scheduled within 2 weeks after 18/10/2026, exact time TBD)

The presentation and the Q&A are graded together as the "meeting" component. You should be able to explain and defend every choice in your project — this is the main way I verify that the work and understanding are your own.

4.1 Presentation checklist

☐ Slides cover motivation, question, dataset/methods, key results, and conclusions/limitations

☐ Fits within 10 minutes when rehearsed (meeting slot is 30 minutes total)

☐ Figures are readable at presentation size (not copy-pasted at tiny resolution)

☐ Slides submitted by the deadline above

5. Using AI (ChatGPT, Claude, etc.)

You are welcome to use AI tools to help write your report, write code, or propose analyses. In all cases, you are responsible for ensuring that everything you submit is correct and relevant.

Your report must include a short AI usage statement covering:

- Which tool(s) you used
- What you used them for (e.g., debugging code, drafting text, suggesting a method)
- The prompts you used

Code or text that is clearly not your own and is not accompanied by prompt information will be considered cheating. You must understand and be able to explain everything in your report and code — during the meeting you may be asked to walk through any part of it.

6. General Requirements

- Submissions are in pairs. Working alone or in a group larger than two requires prior arrangement with the lecturer.
- Be concise. Both the proposal and the final report should include only the information needed to understand what you did and why — avoid padding.
- Your code should be your own. Avoid copying analyses of your dataset found online; your approach should differ from anything already published.
- Project complexity affects the score — overly simple projects will receive a lower grade regardless of correct execution.
- A wider toolkit is rewarded — e.g., combining classification with clustering, comparing several models, or analyzing feature importance — rather than relying on a single method.

7. Grading Rubric

Grades are computed from the three milestones above, broken down as follows. Sub-weights below are percentages of the overall project grade (they sum to 100%).

Component	Sub-criterion	Weight
Proposal (20%)	Question clarity & appropriateness	8%
	Dataset description completeness	6%
	Soundness of proposed methods	6%
Report + Notebook (40%)	Project design (incl. variety/appropriateness of tools used)	8%
	Notebook organization	8%
	Report readability (structure, figure clarity)	8%
	Discussion & interpretation of results	16%

Component	Sub-criterion	Weight
Presentation / Meeting (40%)	Demonstrated understanding of own work	16%
	Clarity of explanation	8%
	Quality of responses to questions	16%

Note: the sub-weights above are a guideline for consistent grading and may be adjusted slightly per project where relevant (e.g., unusually high project complexity).

Appendix A: Presentation & Q&A Grading Guide

This guide breaks the 40-point Presentation/Meeting component into scorable items, for use during or right after each meeting.

Item	Max	What to check	Feedback
Understanding of own work	16	Can explain their own choices (question framing, data prep, method selection) in their own words, without simply reading the slide	
Clarity of explanation	8	Logical flow (question → method → result → conclusion); slides support the talk rather than clutter it; explanation is understandable to a non-specialist	
Quality of responses to questions	16	Answers directly address what was asked; correctly connects answer to their own methods/results; honestly acknowledges a limitation when relevant instead of deflecting	

Scoring anchors

Use these as a quick reference rather than scoring every sub-point separately:

- Full marks — student clearly understands and can defend every choice; answers are direct and accurate; explanation flows logically
- ~75% — solid understanding overall, with one or two spots where the answer is vague, generic, or slightly off-target
- ~50% — surface-level understanding; can describe what they did but struggles to explain why, or gives an off-target answer to a direct question
- Below 50% — cannot explain own choices, or responses suggest the work (or parts of it) were not really understood by the student

Grading rules for this component

- Don't deduct twice for the same gap — if a limitation is missing from the report/notebook and also not raised in Q&A, deduct it once, in whichever rubric item is more directly relevant
- Short, direct answers are fine — do not require a lengthy answer if the student directly and correctly addresses the question
- Qualitative reasoning is acceptable in Q&A if it shows real understanding, even without new numbers or an on-the-spot analysis
- Do not penalize nervousness, accent, or presentation polish — grade content understanding, not delivery style
- If a question goes beyond what a reasonable student could be expected to know (e.g., a very advanced statistical nuance not covered in the course), do not penalize an honest "I don't know" or a reasonable attempt